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Mechanical Properties of Flax Fibre and Polylactic Acid Biocomposites after Accelerated Ageing

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ABSTRACT

Biocomposites based on natural fibres and biodegradable polymers are regarded as an environmentally friendly alternative to glass-fibre plastics in the automotive industry, construction and consumer goods. Their wider use is hindered by sensitivity to moisture and ultraviolet radiation, which reduces strength during service. This study examines the effect of alkaline and silane treatment of flax fibre on the mechanical properties of composites with a polylactic acid matrix after accelerated weathering. Samples with a fibre mass fraction of 30% were produced by hot pressing and subjected to cyclic exposure to ultraviolet radiation, elevated temperature and humidity for 500 and 1000 hours. After 1000 hours of ageing, the tensile strength of composites with untreated fibre decreased by 38%, whereas for composites with silane-treated fibre the decrease was only 17%. Alkaline treatment increased initial strength but had little effect on resistance to ageing. Electron microscopy and infrared spectroscopy showed that silane treatment reduces water uptake and prevents delamination at the fibre-matrix interface. The results support recommending silane treatment for products used outdoors.

Keywords: biocomposites; flax fibre; polylactic acid; ageing; silane treatment

Механические свойства биокомпозитов на основе льняного волокна и полимолочной кислоты после ускоренного старения

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АННОТАЦИЯ

Биокомпозиты на основе натуральных волокон и биоразлагаемых полимеров рассматриваются как экологичная альтернатива стеклопластикам в автомобильной промышленности, строительстве и производстве потребительских товаров. Их широкому применению препятствует чувствительность к влаге и ультрафиолетовому излучению, приводящая к снижению прочности в процессе эксплуатации. В работе исследовано влияние щелочной и силановой обработки льняного волокна на механические свойства композитов с матрицей из полимолочной кислоты после ускоренного климатического старения. Образцы с массовой долей волокна 30 % изготовлены методом горячего прессования и подвергнуты циклическому воздействию ультрафиолета, повышенной температуры и влажности в течение 500 и 1000 часов. После 1000 часов старения прочность при растяжении композитов с необработанным волокном снизилась на 38 %, тогда как для композитов с силанизированным волокном снижение составило лишь 17 %. Щелочная обработка повышала исходную прочность, но слабо влияла на устойчивость к старению. Методами электронной микроскопии и инфракрасной спектроскопии показано, что силановая обработка уменьшает водопоглощение и предотвращает расслоение на границе волокна и матрицы. Полученные результаты позволяют рекомендовать силановую обработку для изделий, эксплуатируемых на открытом воздухе.

Ключевые слова: биокомпозиты; льняное волокно; полимолочная кислота; старение; силановая обработка

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1 INTRODUCTION

Natural fibres — flax, hemp and jute — have low density, adequate specific strength and a renewable raw material base. In combination with biodegradable polymers such as polylactic acid, they make it possible to obtain fully bio-based composites.

The main problem with such materials is the hydrophilicity of the fibre, which impairs adhesion to the hydrophobic matrix and leads to water absorption and degradation in service. The aim of the study is to compare the effectiveness of alkali and silane fibre treatments in improving the resistance of the composites to weathering.

2 MATERIALS AND METHODS

Short flax fibre 3–5 mm long was used. Alkali treatment was carried out with a 5% sodium hydroxide solution for 1 h; silane treatment with a 2% aminosilane solution in a water–ethanol mixture. Composites with a fibre mass fraction of 30% were produced by hot pressing at 185 °C.

Accelerated ageing was carried out in a chamber with 12-h cycles: 8 h of ultraviolet irradiation at 60 °C and 4 h of moisture condensation at 50 °C. Tensile and flexural tests were performed after 0, 500 and 1000 h; at least seven specimens were tested for each condition.

3 RESULTS

The change in tensile strength during ageing is shown in Fig. 1. Initially, the composites with alkali-treated fibre had the highest strength, but after 1000 h their advantage disappeared.

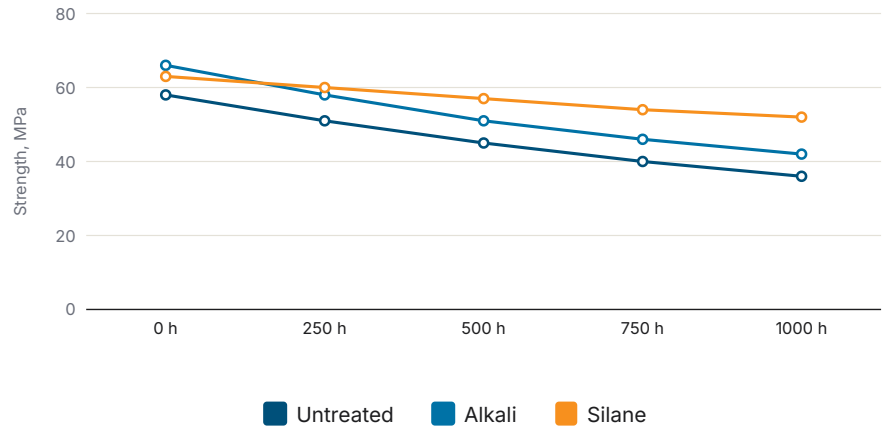


Fig. 1. Tensile strength of the composites as a function of ageing duration, MPa

Summary mechanical properties and water absorption are given in Table 1. Silane treatment reduced water absorption almost by half and ensured the best retention of properties.

Table 1. Properties of the composites before and after 1000 h of ageing

Treatment	Strength, MPa (0 h)	Strength, MPa (1000 h)	Reduction, %	Water absorption, %
Untreated	58	36	38	6,8
Alkali	66	42	36	5,9
Silane	63	52	17	3,5

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4 DISCUSSION

The increase in initial strength after alkali treatment is associated with the removal of hemicelluloses and increased fibre surface roughness, but the hydrophilicity of the fibre is retained. Silane treatment forms a hydrophobic layer on the fibre surface that is chemically bonded to the matrix, which explains the better retention of properties during ageing.

Electron microscopy of the fracture surfaces confirms this: in the untreated composites, fibre pull-out from the matrix is observed after ageing, whereas in the silanised specimens fracture occurs predominantly through the fibres.

5 CONCLUSION

Silane treatment of flax fibre substantially improves the resistance of polylactic acid-based biocomposites to weathering and can be recommended for outdoor products.

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